

feature_matrix_julia_allGO_new.tsv quality analysis

Genes: 602

GO terms: 6,368

Density: 0.0249

Median active terms/gene: 112.5

Median genes/term: 8.0

Singleton/rare terms support ≤ 2 : 1,280

Terms support ≤ 5 : 2,583

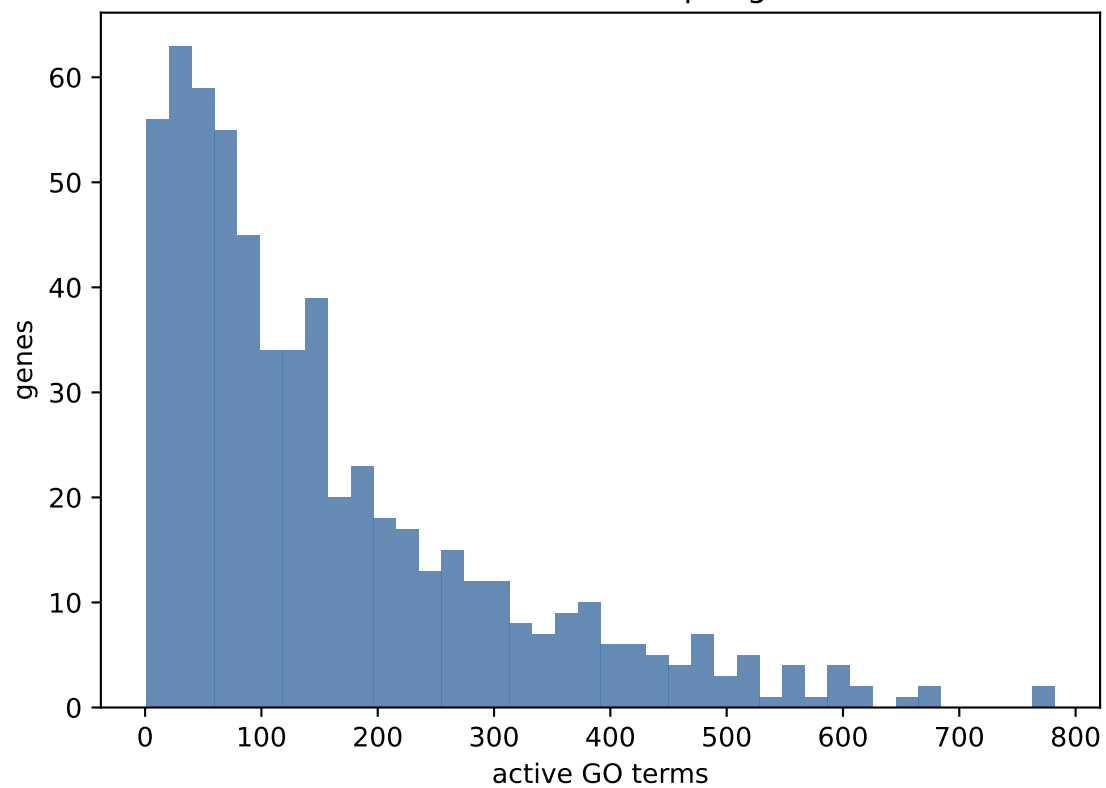
Terms present in $\geq 50\%$ genes: 0

Duplicate gene patterns: 1 groups

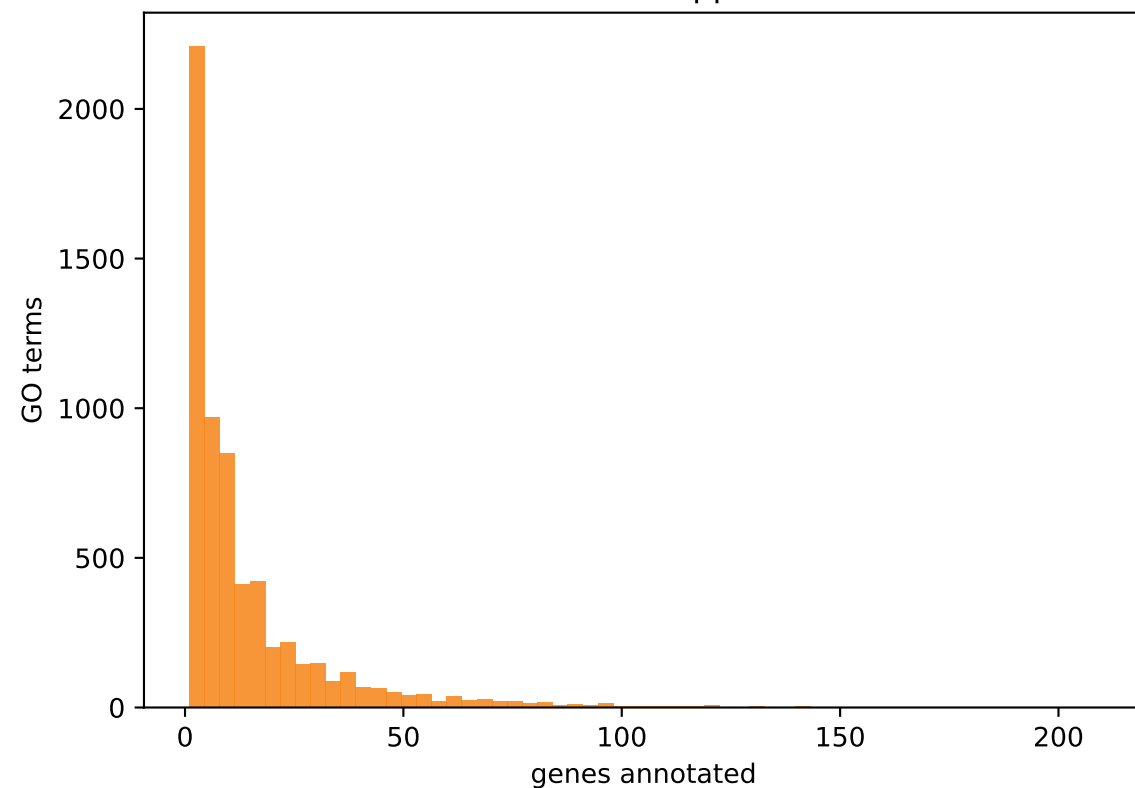
Duplicate GO-term patterns: 459 groups

Interpretation: sparse rare terms increase GO-IC overfitting risk, while broad terms can dominate distance geometry. Use this report to decide filtering/sensitivity grids.

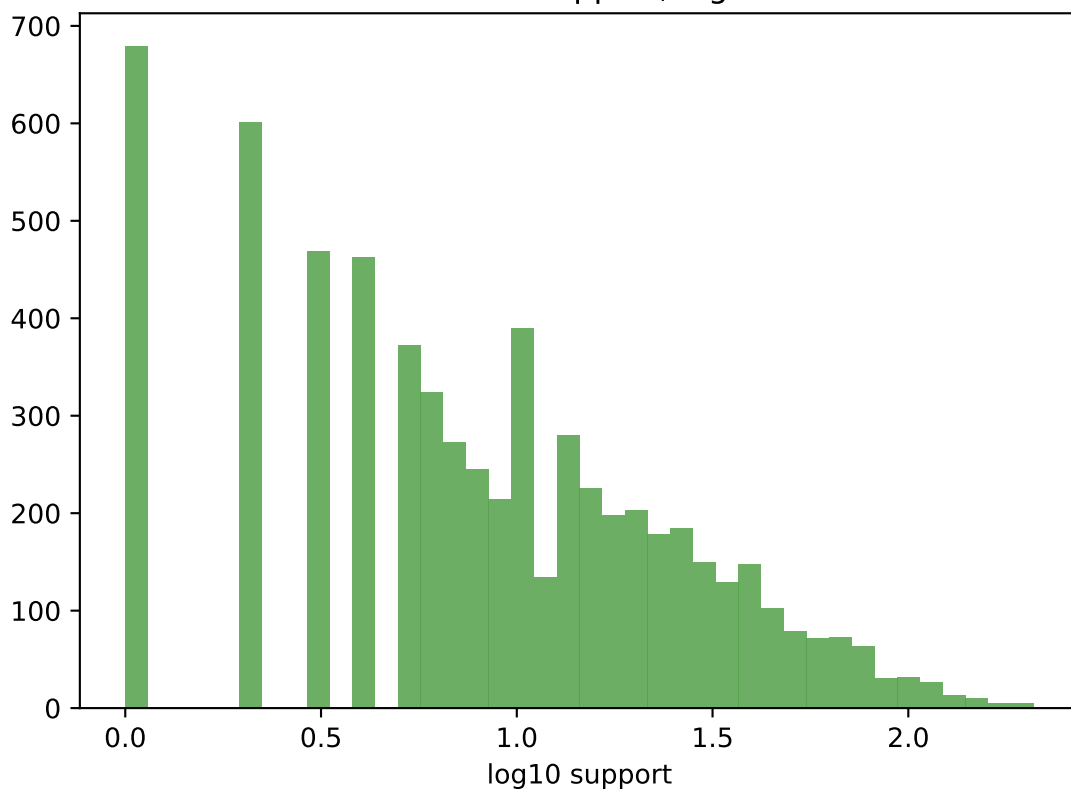
Annotation burden per gene



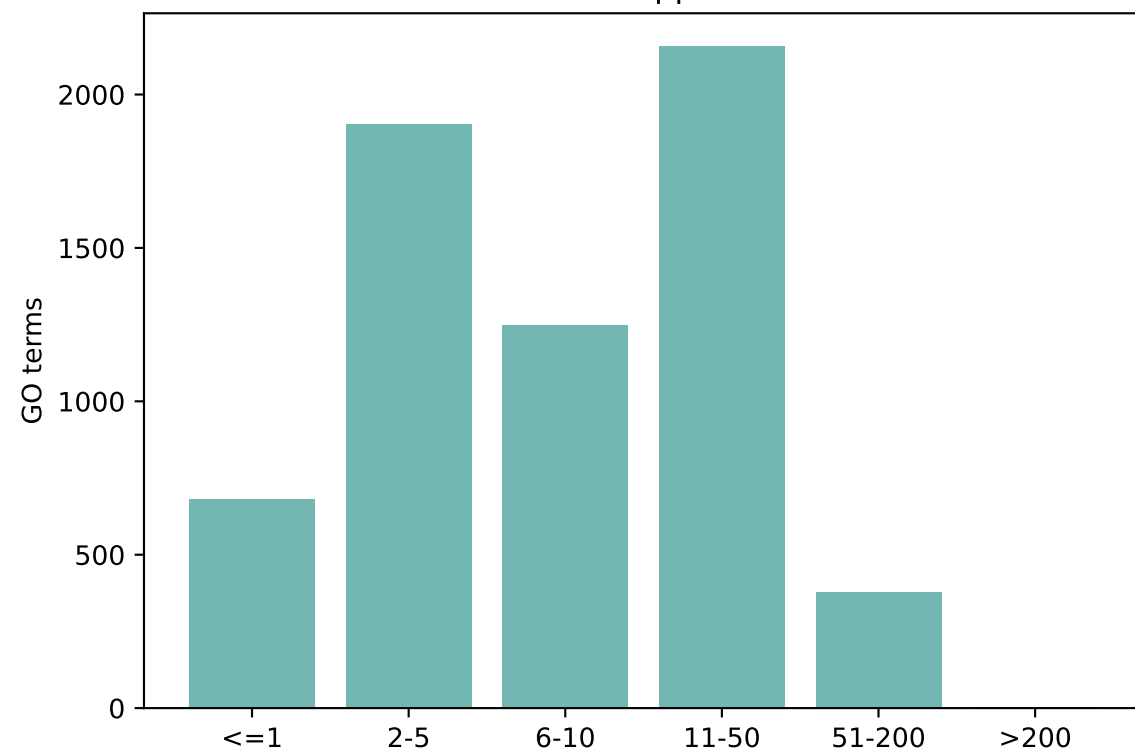
GO term support



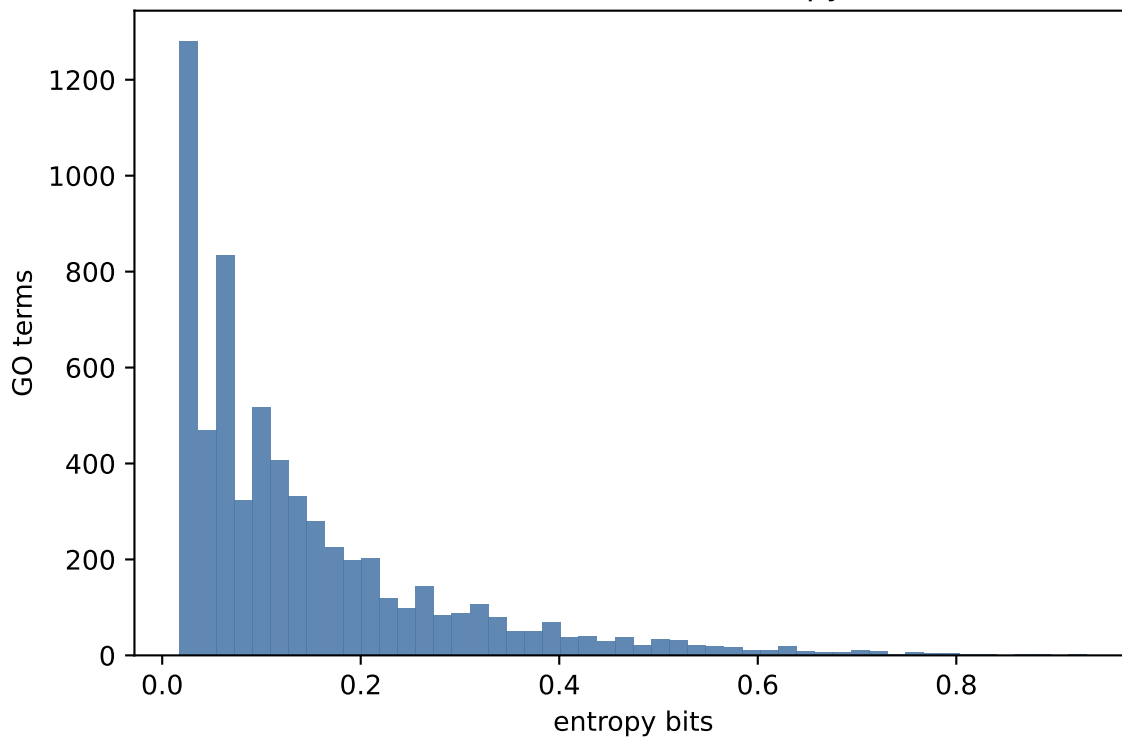
GO term support, log10



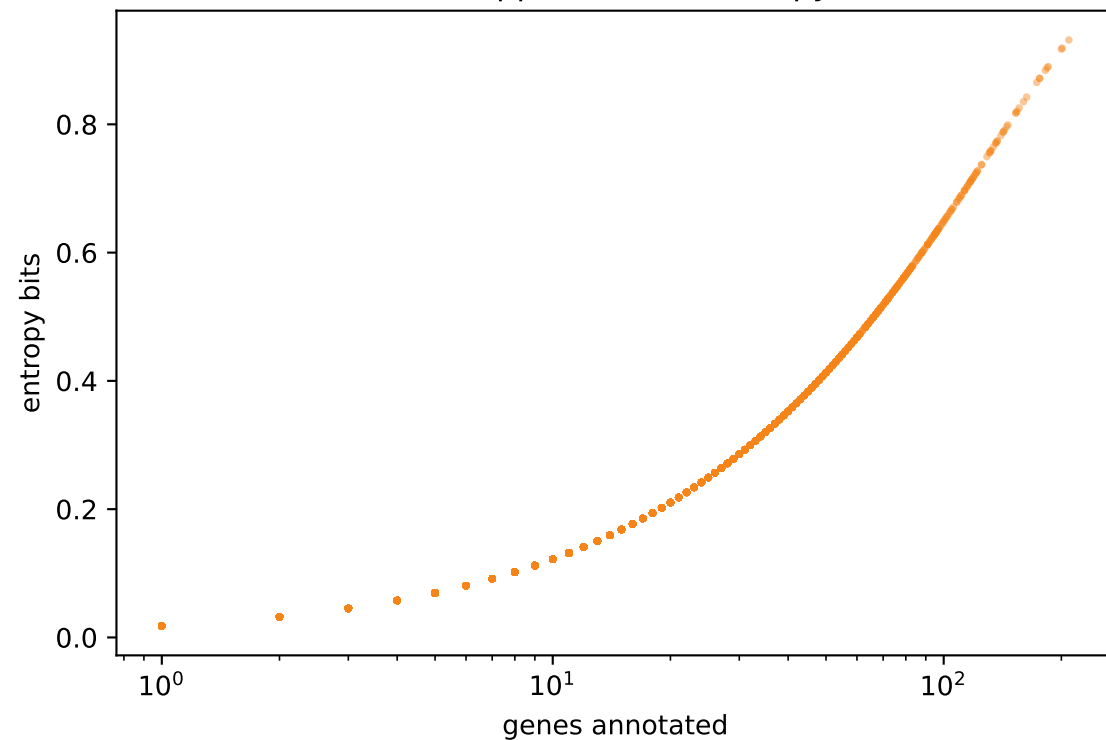
GO term support bins



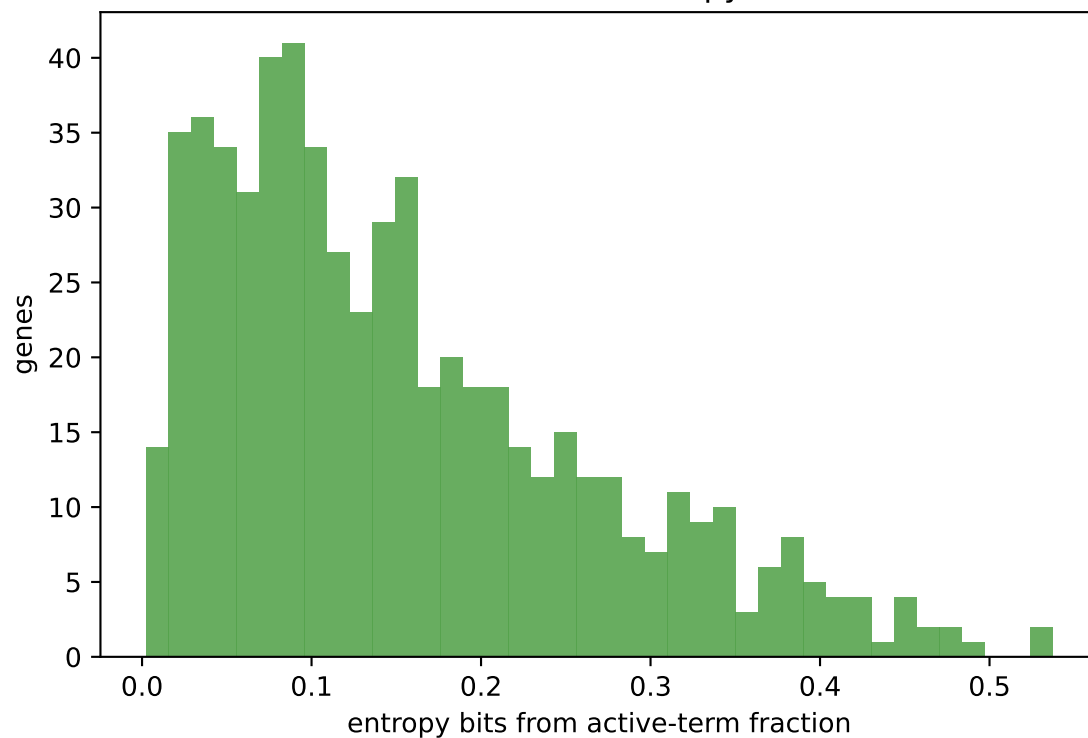
GO-term Bernoulli entropy



GO support versus entropy



Gene row entropy

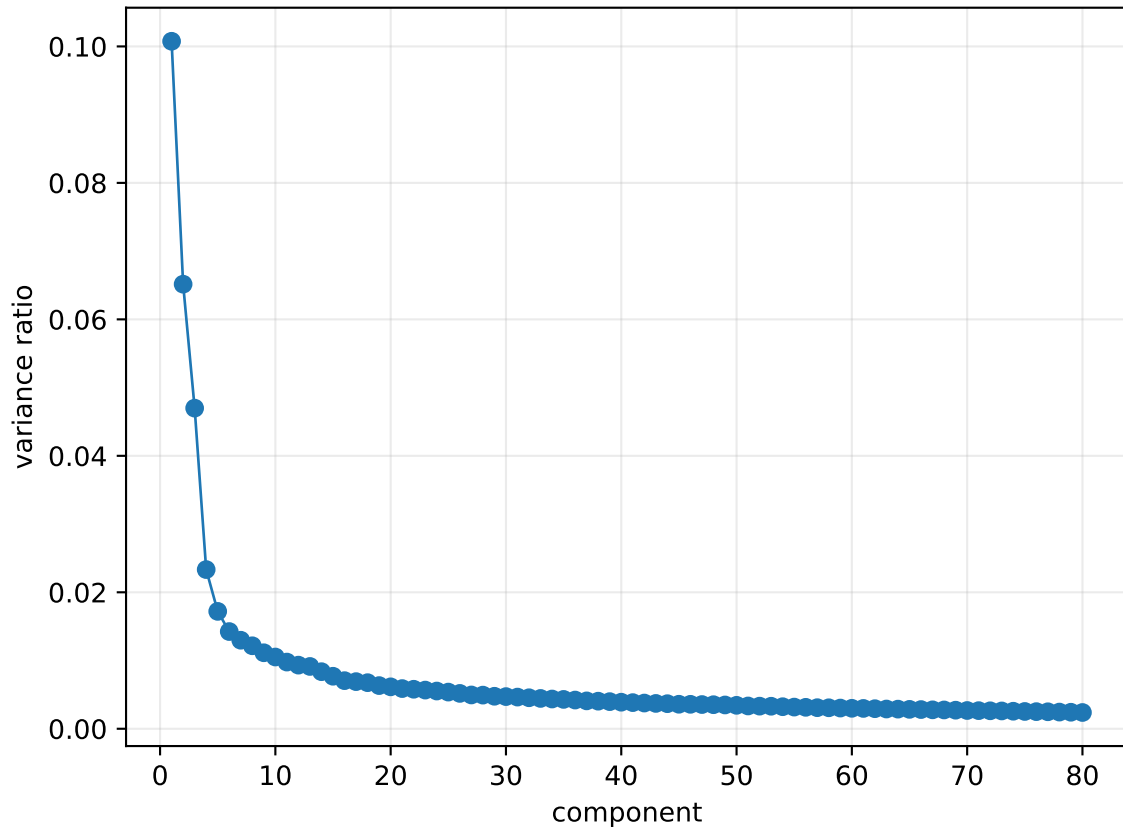


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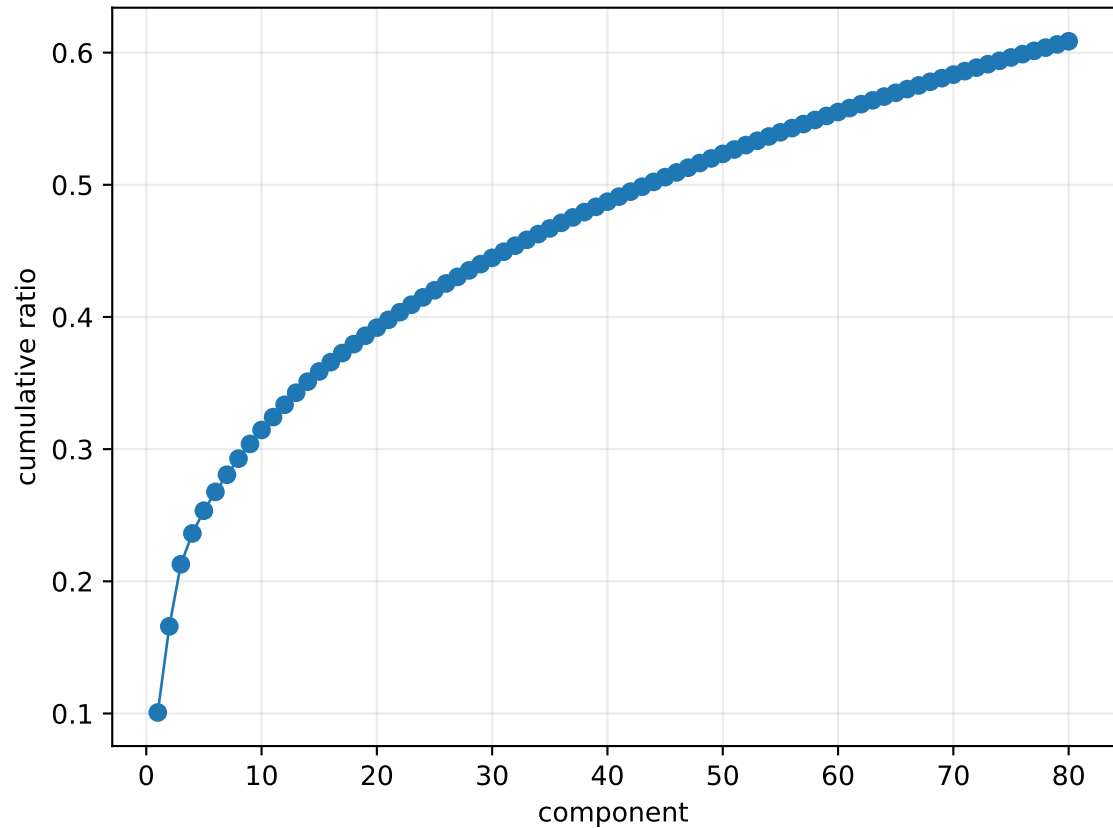
Entropy summary
go_term_entropy_bits_mean: 0.144802
go_term_entropy_bits_median: 0.101883
go_term_entropy_bits_p25: 0.0452878
go_term_entropy_bits_p75: 0.185483
go_term_entropy_bits_p95: 0.441
go_term_entropy_bits_max: 0.931521
go_terms_entropy_le_0_05: 1749
go_terms_entropy_le_0_10: 3180
go_terms_entropy_le_0_20: 4866
go_terms_entropy_ge_0_50: 223
go_terms_entropy_ge_0_80: 15
total_go_marginal_entropy_bits: 922.099
mean_go_marginal_entropy_bits_per_term: 0.144802
matrix_density_entropy_bits_per_cell: 0.168274
gene_row_entropy_bits_mean: 0.154398
gene_row_entropy_bits_median: 0.12813
gene_row_entropy_bits_p95: 0.381631
gene_row_entropy_bits_max: 0.53736
annotation_entry_entropy_across_genes_bits: 8.69263
annotation_entry_effective_genes: 413.755
annotation_entry_entropy_across_go_terms_bits: 11.732
annotation_entry_effective_go_terms: 3401.59

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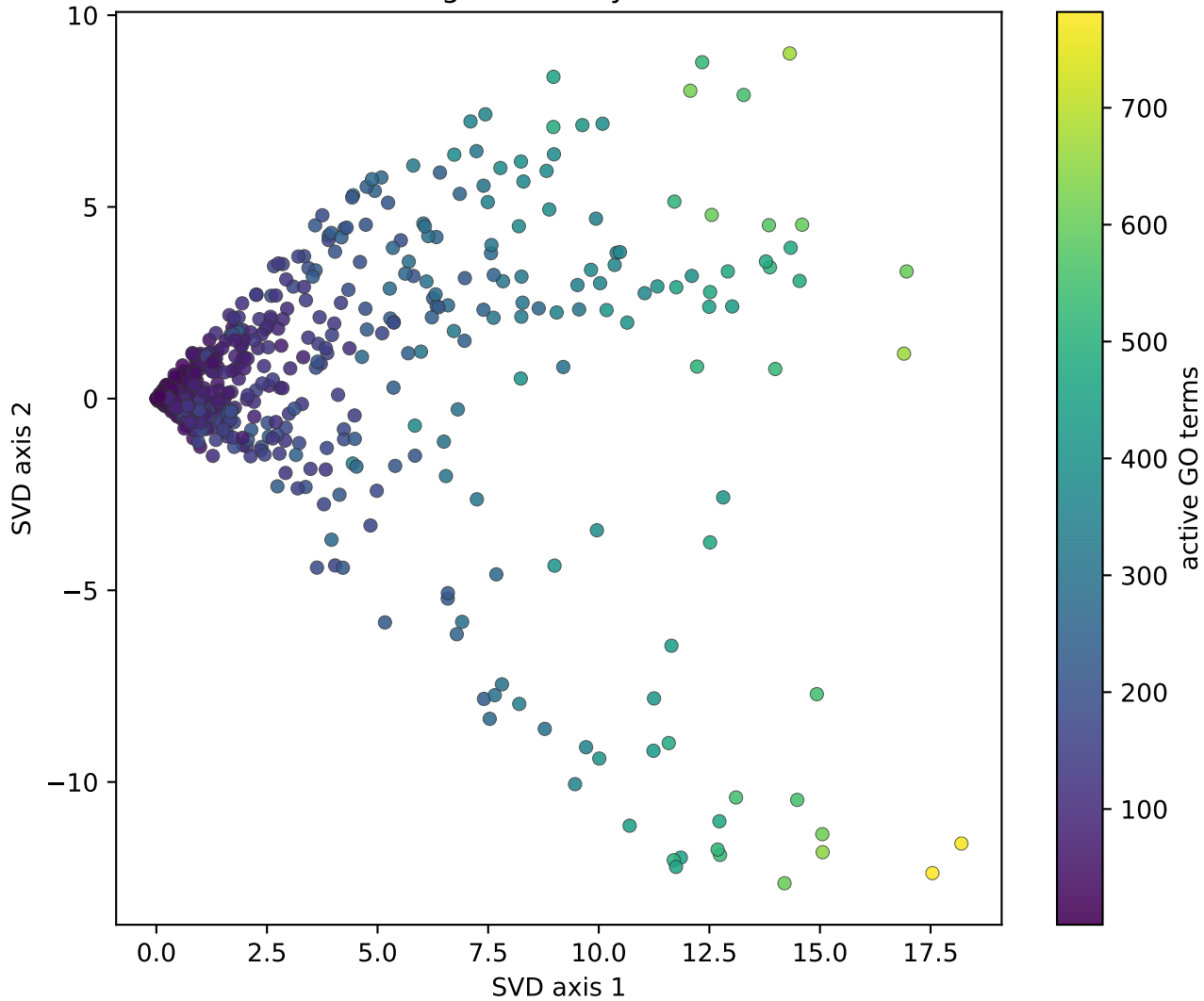
SVD explained variance ratio



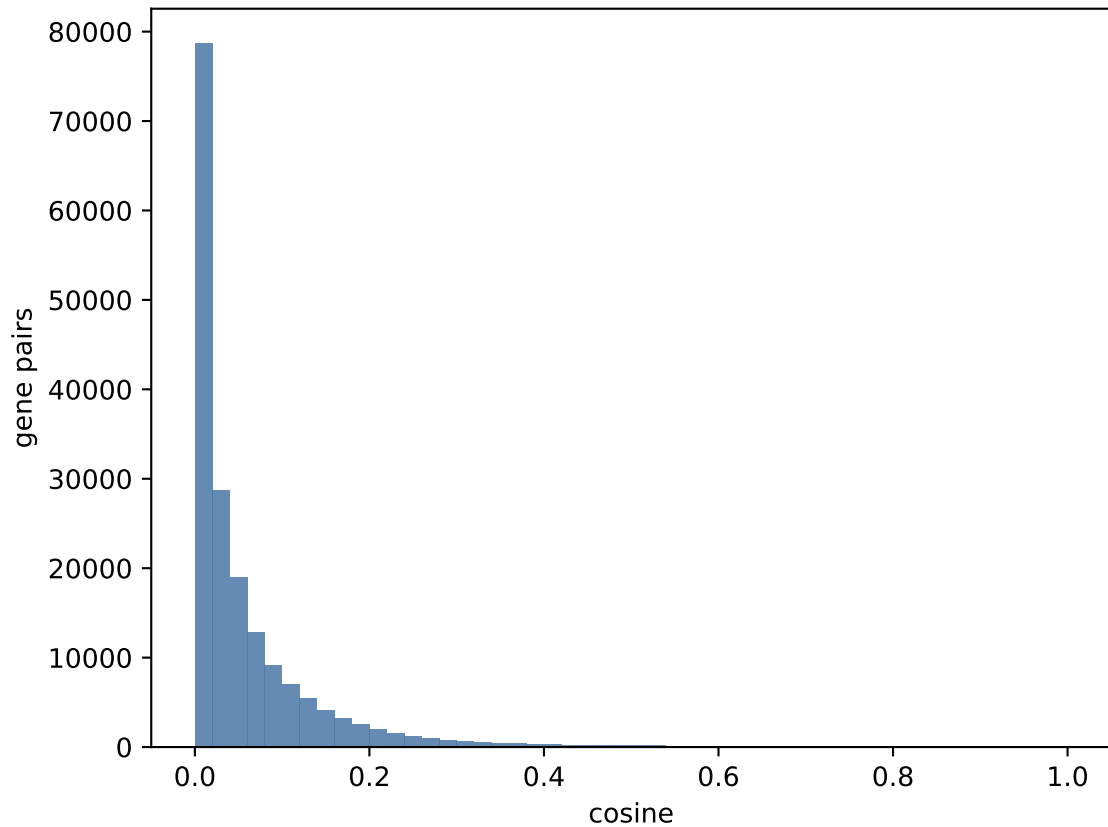
Cumulative SVD variance



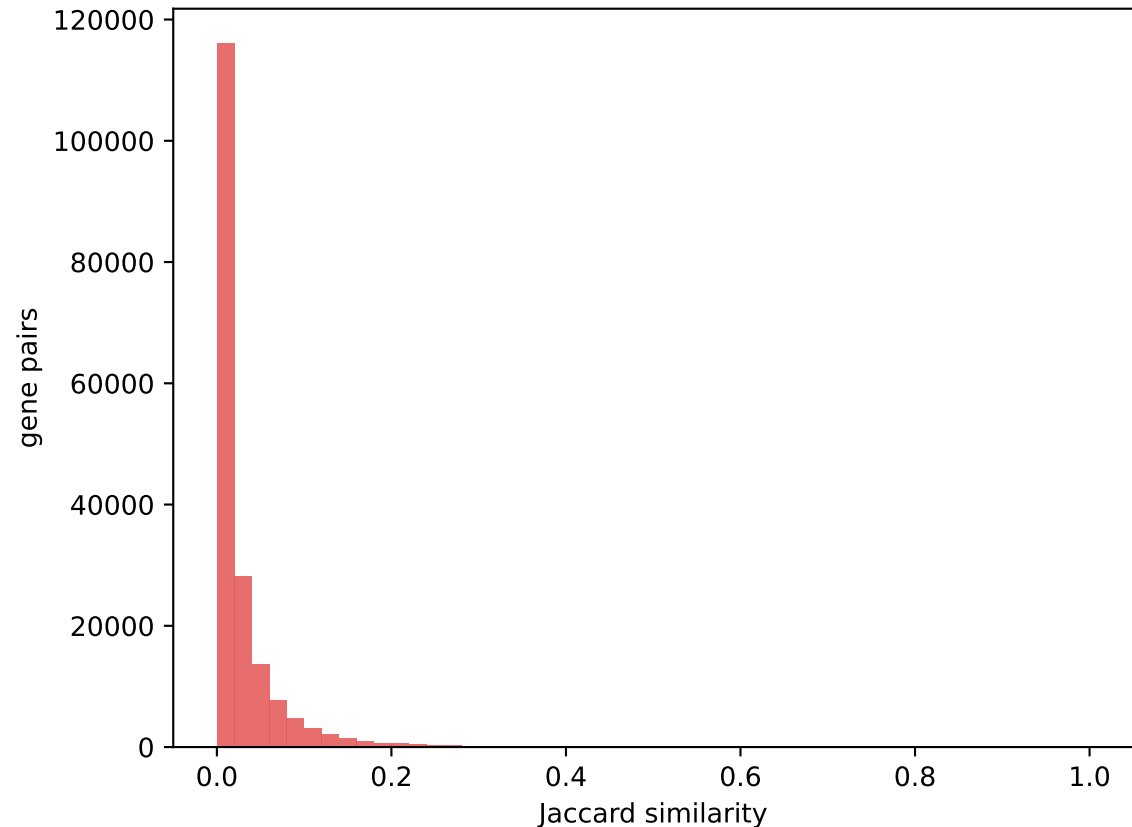
Gene embedding colored by annotation burden



Pairwise gene cosine similarity



Pairwise gene Jaccard similarity



GO terms retained under support/prevalence filters

